

Experiment 15

Aim: To demonstrate the segmentation of image using python.

Program

```
import cv2

import numpy as np

from matplotlib import pyplot as plt

# from google.colab import drive

# Mount Google Drive

# drive.mount('/content/drive')

# For Colab: !wget https://example.com/cataract_image.jpg -O image.jpg

# (replace with actual URL)


# --- New code to download image from URL ---

# Example URL for a dog image. You can replace this with any public image URL.

!wget -O downloaded_image.jpg
https://cdn.pixabay.com/photo/2016/12/13/05/15/puppy-1903313_1280.jpg

img = cv2.imread('downloaded_image.jpg') # Changed path to the downloaded image

# -----

# Check if the image was loaded successfully

if img is None:

    print("Error: Could not load image. Please check the file path.")

else:

    rgb_img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    pixels = np.float32(rgb_img.reshape((-1, 3)))

    criteria = (cv2.TERM_CRITERIA_EPS +

                cv2.TERM_CRITERIA_MAX_ITER, 100, 0.2)

    K = 3
```

```

_, labels, centers = cv2.kmeans(pixels, K, None, criteria, 10,
                                cv2.KMEANS_RANDOM_CENTERS)

centers = np.uint8(centers)

segmented_img = centers[labels.flatten()].reshape(rgb_img.shape)

plt.figure(figsize=(10, 5))

plt.subplot(121)

plt.imshow(rgb_img)

plt.title('Original Image')

plt.axis('off')


plt.subplot(122)

plt.imshow(segmented_img)

plt.title('Segmented Image (K-means)')

plt.axis('off')

plt.tight_layout()

plt.show()

```

Output:

